

Efficient UAV-based 3D Reconstruction through Online Quality Assessment and Adaptive Re-Scanning

Sanggi Lee^a and *Soohwan Song^b

^a Department of Computer Science and Artificial Intelligence, Dongguk University, Seoul, Korea,
sglee@dgu.ac.kr

^b Department of Computer Science and Artificial Intelligence, Dongguk University, Seoul, Korea,
songsh@dongguk.edu

Abstract

Conventional UAV-based 3D scanning methods suffer from inefficiencies in offline approaches and the heavy resource demands of online methods. This paper introduces an adaptive re-scanning path planning framework that leverages real-time multi-view stereo reconstruction on a ground server during the UAV's initial flight. As the preliminary 3D model is built, low-quality regions are detected on the fly, and a re-scanning path is generated for execution during the return flight, improving completeness without requiring a second mission. Experiments in a high-fidelity simulation show that unobserved regions were reduced from 31.48% to 22.82%, and incomplete areas dropped from 6.63% to 0.03%, confirming the effectiveness of the proposed method. This framework offers a practical solution for producing accurate 3D models within a single autonomous flight.

Keywords: View path planning, 3D reconstruction, coverage path planning, multi-view stereo

1. Introduction

Unmanned Aerial Vehicles (UAVs), commonly referred to as drones, have become essential tools for 3D reconstruction in a wide range of applications, including topographic surveying, infrastructure inspection, cultural heritage preservation, virtual reality content creation, and digital twin development [1]. By capturing high-resolution, overlapping images

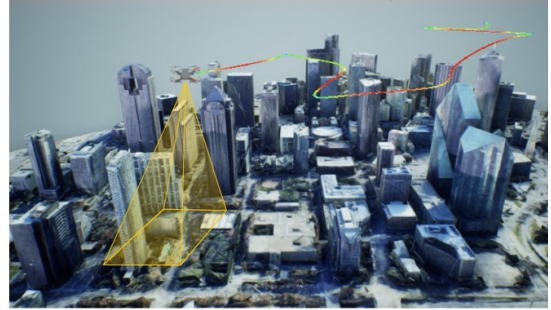


Figure 1: Illustration of the proposed method applied to 3D scanning in a complex urban environment. Our method performs the initial scan while the UAV follows a coverage path toward the destination. Simultaneously, it generates a re-scanning path for the return trip from the destination back to the starting point, targeting regions with incomplete reconstruction. This strategy enables the creation of a complete and high-quality 3D model in a single flight.

from multiple viewpoints, UAVs enable the efficient and cost-effective collection of spatial data necessary for generating detailed and geometrically accurate 3D models.

The quality and completeness of these reconstructions depend heavily on the data acquisition strategy, particularly the flight path taken during image collection. Many existing UAV-based 3D mapping methods adopt an offline reconstruction approach [2,3]. In this paradigm, images captured during an initial flight are first processed to generate a preliminary 3D model. Areas with poor reconstruction quality or insufficient coverage are then identified, and the UAV is re-deployed to revisit those regions. While this approach allows for targeted improvement, it lacks real-time adaptability and often

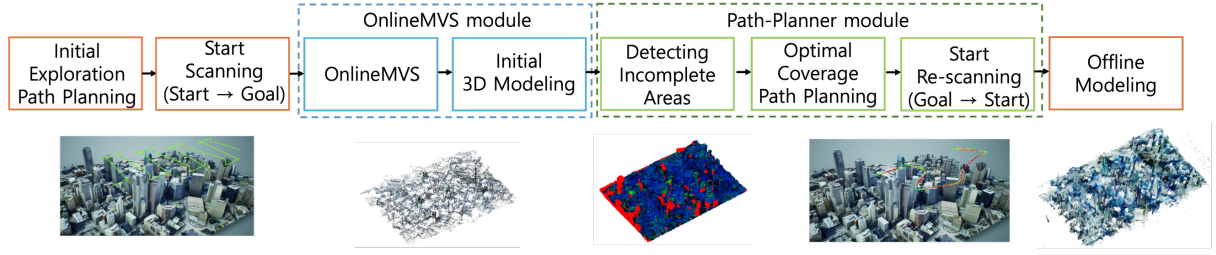


Figure 2: The proposed system comprises an end-to-end workflow that integrates initial path planning, online 3D reconstruction, and adaptive re-exploration. During the initial flight, the UAV follows a predefined trajectory while the online MVS module incrementally generates a 3D point cloud in real time. Concurrently, the system evaluates reconstruction quality to identify unobserved or poorly scanned regions. Based on this online assessment, a re-exploration path is dynamically computed and executed during the UAV's return flight. This integrated process culminates in the generation of a refined and geometrically complete 3D model, achieved within a single flight mission.

requires additional flight missions, resulting in increased operational time and cost.

To overcome these drawbacks, online reconstruction methods have been explored [4, 5]. These systems update the UAV's flight path in real time based on live reconstruction feedback, allowing for adaptive trajectory refinement during flight. Although promising in theory, online methods introduce significant challenges. They require continuous communication, low-latency data transmission, and high onboard computational power. Moreover, because the UAV must frequently adjust its trajectory based on dynamic feedback, flight stability may be compromised. These constraints are difficult to meet with the limited hardware and energy resources typically available on UAV platforms, limiting the practicality of such systems in real-world deployments.

To address these limitations, this study proposes an adaptive re-scanning path planning strategy that blends the strengths of both offline and online approaches. In the proposed framework, online reconstruction is conducted concurrently during the UAV's initial flight. As the UAV collects image data, the system continuously analyzes reconstruction quality and identifies regions that are incomplete or of low fidelity. Based on this information, an additional re-scanning path is generated on-the-fly, allowing the UAV to revisit problematic areas during its return to the original starting point, without requiring a landing or a separate re-flight. This approach significantly reduces mission overhead while improving the geometric completeness and accuracy of the final 3D model. It offers a practical and efficient solution for UAV-based 3D mapping tasks, particularly in scenarios where both reconstruction quality and operational efficiency are critical.

2. Proposed Method

We present an adaptive UAV-based 3D scanning framework that integrates online multi-view stereo (MVS) reconstruction with real-time re-scanning path planning. During the outbound flight, the UAV follows a predefined trajectory to cover the target area. In parallel, a ground server performs online MVS reconstruction to generate an initial 3D point cloud and detect under-scanned regions. Based on this reconstruction feedback, a re-scanning path is computed on the fly. This path allows the UAV to revisit incomplete regions during its return flight—without requiring a landing or re-launch. As a result, high-quality 3D data can be acquired within a single flight. After the mission, the captured images are processed offline to generate a high-resolution textured model. The full pipeline is illustrated in Figure 2.

1) Online MVS for Initial Reconstruction: The online MVS module [12] incrementally constructs a 3D point cloud from incoming image streams during flight. Camera poses are estimated using a hybrid visual SLAM method that combines direct optical flow tracking with global bundle adjustment, inspired by DROID-SLAM [6]. A dynamic keyframe graph is maintained, where nodes represent keyframes and edges encode co-visibility relationships.

Depth estimation is performed using a cascade cost-volume MVS network [7], which refines depth across multiple stages. This design reduces GPU memory consumption and enables near real-time inference. To ensure geometric consistency, initial depth maps are initialized from SLAM-based disparities. Low-confidence depth values are filtered out, and only those consistent with neighboring views are retained.

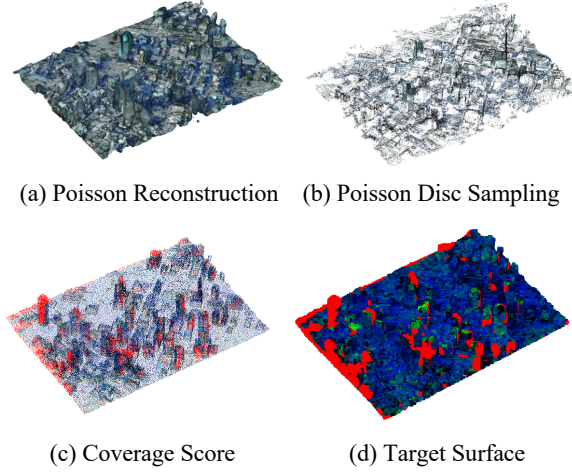


Figure 3: The quality assessment pipeline includes (a) Poisson surface reconstruction from the initial point cloud, (b) uniform point sampling via Poisson disc sampling, (c) computation of per-point coverage scores, and (d) extraction of low-score regions as the final target surface for re-scanning. Uncovered regions are highlighted in red, while low-score regions are highlighted in green.

Every 30 keyframes, global bundle adjustment is applied to jointly optimize camera poses and the 3D structure. The global point cloud is then updated using rigid transformations, preserving accuracy while avoiding non-rigid deformations. This process ensures a geometrically consistent and precise 3D reconstruction during flight.

2) Surface Model Construction for Re-Scanning Analysis: As the UAV moves toward the destination, the generated point cloud is refined and transformed into a surface model for quality evaluation. This enables accurate identification of under-scanned regions. As shown in Figure 3, Poisson Surface Reconstruction [8] is used to generate a continuous surface mesh using estimated surface normal. Next, Poisson Disc Sampling is applied to extract uniformly distributed surface points, eliminating redundancy and ensuring spatial consistency.

Each sampled surface point serves as a unit for evaluating reconstruction completeness. A coverage score [9] is computed for every point. Regions with low scores, visualized in red and green in Figure 3, are marked as target areas for re-scanning during the return flight.

3) Coverage Score Computation: To assess surface completeness, a coverage score is assigned to each sampled point. The score ranges from 0 to 1,

where 0 indicates insufficient observation and 1 indicates complete and high-quality coverage.

The final score is computed as the product of three components: distance, incidence angle, and view angle scores :

$$S_{final} = S_{dist} \times S_{incidence} \times S_{view_angle}$$

where S_{dist} is the distance score penalizes viewpoints that are too far from the target surface, defined as:

$$S_{dist} = \max(0.0, 1.0 - \frac{d}{d_{max}})$$

where d is the distance to the target point and d_{max} is the maximum effective sensing distance. The incidence angle score, $S_{incidence}$, favors viewpoints that are perpendicular to the target surface for better scanning quality:

$$S_{incidence} = \max(0.0, -v_{view} \cdot n_{surface})$$

where v_{view} is the view vector and $n_{surface}$ is the surface normal vector. The view angle score, S_{view_angle} , gives a higher score to targets closer to the center of the camera's field of view:

$$S_{view_angle} = \max(0.0, 1.0 - \frac{\theta}{\theta_{fov_base}})$$

where θ is the angle between the camera's principal axis and the target. By combining these factors, the score robustly evaluates how well each region has been captured, facilitating the identification of regions that need re-scanning.

4) Re-Scanning Path Planning via Sampling-Based Coverage Strategy: To efficiently revisit under-scanned regions, the system employs a Sampling-Based Coverage Path Planning (SCPP) approach [10]. Initially, candidate viewpoints are randomly sampled on the target surface. For each viewpoint, the system analyzes visibility and computes information gain, defined as the expected improvement in coverage score after scanning from that position.

Viewpoints with high information gain are selected first. Additional viewpoints are incrementally added until all target regions are sufficiently covered. Once the final set of viewpoints is determined, a Traveling Salesman Problem (TSP) solver [11] is used to generate the shortest path that visits all points. The resulting path is smoothed to eliminate sharp turns, improving both flight stability and mission efficiency.

The smoothed path is then converted into a polynomial trajectory that considers UAV dynamics and kinematic constraints. This trajectory is

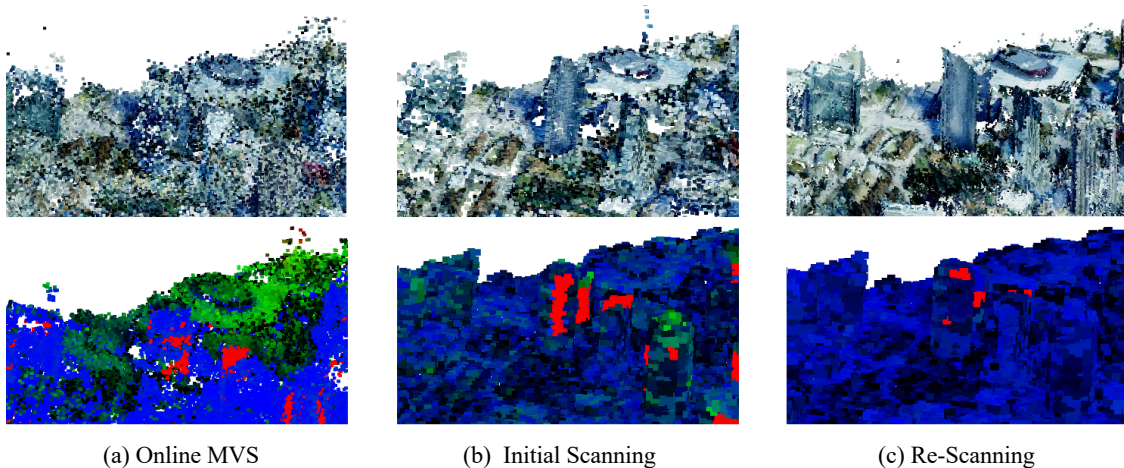


Figure 4: Qualitative results of the proposed method. (a) Online MVS reconstruction, (b) after initial scanning, and (c) after re-scanning. The bottom row shows quality maps, with red indicating unscanned and green indicating low-coverage regions, both significantly reduced after re-scanning.

Method	Initial Scanning	Re-Scanning
Uncovered Regions rate (%)	31.48	22.82
Incomplete Regions rate (%)	6.63	0.03

Table 1: Quantitative results showing a reduction in unobserved (31.48% $\rightarrow$ 22.82%) and incomplete regions (6.63% $\rightarrow$ 0.03%) after re-scanning, confirming improved model completeness with the proposed method.

transmitted to the flight controller, enabling the UAV to conduct high-precision re-scanning without unnecessary maneuvers. The result is a complete and geometrically accurate 3D model acquired within a single, efficient flight mission.

3. Experimental Results

To evaluate the performance of the proposed autonomous 3D reconstruction system, we conducted experiments in a high-fidelity simulation environment. This section outlines the experimental setup, procedures, and both qualitative and quantitative evaluation results.

All experiments were performed in a photorealistic 3D urban environment developed using Unreal Engine 4 (UE4). UE4 provides realistic rendering and a physics engine, offering a reliable testbed for validating autonomous algorithms under conditions that closely resemble the real world. For UAV and sensor simulation, we employed the AirSim plugin, which supports accurate modeling of drone

dynamics and enables image acquisition via a virtual RGB camera.

The simulation system was integrated with the Robot Operating System (ROS). During flight, image streams and drone state data were transmitted in real time to the online MVS and path planning modules via ROS topics. Once a re-exploration path was computed, it was sent back to the UAV’s flight controller to execute autonomous re-scanning.

To validate the effectiveness of the proposed method, we compared the reconstructed 3D models before and after re-scanning against a ground-truth (GT) model. Figure 4 shows the qualitative comparison results. As illustrated in Figure 4b, the initial flight leaves substantial portions of the environment unobserved (highlighted in red) and results in large incomplete surface regions (in green), especially in occluded or oblique-view areas. In contrast, Figure 4c shows that after applying the proposed re-scanning strategy, both unobserved and incomplete regions are significantly reduced. This visual improvement confirms that the re-exploration process enhances overall model completeness and geometric coverage.

To objectively measure the improvement, we calculated the proportion of Unobserved Regions and Incomplete Regions before and after re-scanning. The quantitative results are summarized in Table 1. After the initial scan, unobserved regions accounted for 31.48%, and incomplete regions for 6.63% of the total surface. Following re-exploration, the unobserved region rate dropped to 22.82%, and the incomplete region rate decreased sharply to 0.03%. These results clearly demonstrate that the proposed

re-scanning path planning algorithm significantly improves 3D reconstruction quality by enhancing coverage and surface completeness.

4. Conclusion

This paper proposed an adaptive path planning framework for UAV-based 3D reconstruction that enables efficient re-exploration of poorly observed regions without overloading onboard computation. The method uses online MVS during the initial flight to generate a preliminary model and immediately detects low-quality areas. A re-scanning path is then computed and executed within the same mission, improving reconstruction without additional flights. Experiments in a realistic urban simulation show that the framework enhances model completeness and geometric fidelity over fixed-path methods. Results confirm its effectiveness in reducing unobserved and incomplete regions. Future work includes real-world UAV deployment, integration of advanced quality metrics, and occlusion-aware strategies for complex environments.

Acknowledgments

This research was supported in part by the Institute for Information & Communications Technology Planning & Evaluation (IITP) through the ICT Challenge and Advanced Network of Human Resource Development (ICAN-HRD) program, funded by the Korean Government (Ministry of Science and ICT) under Grant No. IITP-2025-RS-2023-00260248, and in part by the Artificial Intelligence Convergence Innovation Human Resources Development Program under Grant No. IITP-2025-RS-2023-00254592, also supervised by the IITP.

References

- [1] Colomina, I., & Molina, P. (2014). Unmanned aerial systems for photogrammetry and remote sensing: A review. *ISPRS Journal of photogrammetry and remote sensing*, 92, 79-97.
- [2] Hepp, B., Nießner, M., & Hilliges, O. (2018). Plan3d: Viewpoint and trajectory optimization for aerial multi-view stereo reconstruction. *ACM Transactions on Graphics (TOG)*, 38(1), 1-17.
- [3] Feng, C., Li, H., Zhang, M., Chen, X., Zhou, B., & Shen, S. (2024, May). Fc-planner: A skeleton-guided planning framework for fast aerial coverage of complex 3d scenes. In *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 8686-8692). IEEE.
- [4] Song, S., Kim, D., & Choi, S. (2021). View path planning via online multiview stereo for 3-d modeling of large-scale structures. *IEEE Transactions on Robotics*, 38(1), 372-390.
- [5] Kuang, Q., Wu, J., Pan, J., & Zhou, B. (2020, May). Real-time UAV path planning for autonomous urban scene reconstruction. In *2020 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 1156-1162). IEEE.
- [6] Teed, Z., & Deng, J. (2021). Droid-slam: Deep visual slam for monocular, stereo, and rgb-d cameras. *Advances in neural information processing systems*, 34, 16558-16569.
- [7] Gu, X., Fan, Z., Zhu, S., Dai, Z., Tan, F., & Tan, P. (2020). Cascade cost volume for high-resolution multi-view stereo and stereo matching. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 2495-2504).
- [8] Kazhdan, M., & Hoppe, H. (2013). Screened poisson surface reconstruction. *ACM Transactions on Graphics (ToG)*, 32(3), 1-13.
- [9] Maboudi, M., Homaei, M., Song, S., Malihi, S., Saadatseresht, M., & Gerke, M. (2023). A review on viewpoints and path planning for UAV-based 3-D reconstruction. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 16, 5026-5048.
- [10] Englot, B., & Hover, F. (2012, May). Sampling-based coverage path planning for inspection of complex structures. In *Proceedings of the International Conference on Automated Planning and Scheduling* (Vol. 22, pp. 29-37).
- [11] Dantzig, G., Fulkerson, R., & Johnson, S. (1954). Solution of a large-scale traveling-salesman problem. *Journal of the operations research society of America*, 2(4), 393-410.
- [12] Lee, Byeonggwon, et al., Online 3D gaussian splatting modeling with novel view selection., *IJCAI* 2025.